

MARKING SCHEME MAIN EXAM OF ADVANCED CALCULUS FOR ENG

SECTION A. ATTEMPT ALL QUESTIONS /20 marks**Question one.**

- a) Find the Wronskian of the set $\{x, \cos -x, 1\}$ and state if the set of solutions is independent or not **(2 marks)**

- b) The Laplace transform of a function $f(t)$ is given as $F(s) = \frac{2s+1}{s(s+1)}$. State the Laplace transform of $e^{-2t}f(t)$. **(3 marks)**

- c) Find the $f(t)$ such that $\mathcal{L}\{f(t)\}$ is given as $F(s) = \frac{2}{s^2+4s+5}$ **(2 marks)**

- d) Solve the differential equation
 $y'' - y = -t^2, \quad y(0) = 2; y'(0) = 0.$
 using Laplace transforms. **(3 marks)**

- e) If $\phi(x, y, z) = x^2yz$ and $\vec{A} = 3x^2y\vec{i} + yz^2\vec{j} - xz\vec{k}$, then find $\frac{\partial^2}{\partial y \partial z}(\phi\vec{A})$ at the point $(1, -2, -1)$. **(4 marks)**

- f) Evaluate:

$$\int_{x=0}^{x=1} \int_{y=0}^{y=1} \int_{z=0}^{z=1} (x+y+z) dz dy dx. \quad \textbf{(3 marks)}$$

(3 marks)

- g) Verify that $u(x, y) = xy + \frac{x}{y}$ is a solution of the equation **(3 marks)**

$$y \frac{\partial^2 u}{\partial y^2} + 2x \frac{\partial^2 u}{\partial x \partial y} = 2x$$

Answers:

a) $W = \begin{vmatrix} x & \cos -x & 1 \\ 1 & \sin -x & 0 \\ 0 & -\cos -x & 0 \end{vmatrix} = \begin{vmatrix} 1 & -\sin x \\ 0 & -\cos x \end{vmatrix} \quad \textbf{/0.5 marks}$

$$= -\cos x \quad \textbf{/0.5 marks}$$

$W \neq 0 \Leftrightarrow \cos x \neq 0 \Rightarrow x \neq \pm\left(\frac{\pi}{2} + k\pi\right), \forall k \in \mathbb{Z}$, the set of solutions is independent and otherwise is not independent **/1 mark**

b) Use the first shift theorem as : $\mathcal{L}\{e^{-2t}f(t)\} = F(s+2)$ **/1 mark**

$$F(s+2) = \frac{2(s+2)+1}{(s+2)(s+2+1)}$$

$$= \frac{2s+5}{(s+2)(s+3)}$$

/1 mark**/1 mark**

c) $F(s) = \frac{2}{s^2+4s+5} = \frac{2}{(s+2)^2+1} = 2 \times \frac{1}{(s+2)^2+1}$

$$= \mathcal{L}\{2e^{-2t}\sin t\}$$

/1 mark**/0.5 marks**

Hence $f(t) = 2e^{-2t}\sin t$ **/0.5 marks**

d) Let $\mathcal{L}\{y(t)\} = Y(s) = Y$

$$\mathcal{L}\{y'' - y\} = \mathcal{L}\{-t^2\} \Leftrightarrow \mathcal{L}\{y''\} - \mathcal{L}\{y\} = \mathcal{L}\{-t^2\}$$

/0.5 marks

$$\Leftrightarrow s^2Y - sy(0) - y'(0) - Y = \frac{2}{s^3} \Leftrightarrow s^2Y - 2s - Y = -\frac{2}{s^3}$$

/1 mark

$$\Rightarrow Y = \frac{2(s^2+1)}{s^3} = \frac{2}{s} + \frac{2}{s^3}$$

/0.5 marks

Taking the inverse Laplace transform gives

$$y(t) = 2 + t^2$$

/1 mark

e) $\phi\vec{A} = x^2yz(3x^2y\vec{i} + yz^2\vec{j} - xz\vec{k}) = 3x^4y^2z\vec{i} + x^2y^2z^3\vec{j} - x^3yz^2\vec{k}$ **/1 mark**

$$\frac{\partial}{\partial z}(\phi\vec{A}) = 3x^4y^2\vec{i} + 3x^2y^2z^2\vec{j} - 2x^3yz\vec{k}$$

/1 mark

$$\frac{\partial^2}{\partial y \partial z}(\phi\vec{A}) = 6x^4y\vec{i} + 6x^2yz\vec{j} - 2x^3z\vec{k}$$

/1 mark

$$\frac{\partial^2}{\partial y \partial z}(\phi\vec{A})\Big|_{(1,-2,-1)} = -12\vec{i} - 12\vec{j} + 2\vec{k}$$

/1 mark

f) $\int_{x=0}^{x=1} \int_{y=0}^{y=1} \int_{z=0}^{z=1} (x+y+z) dz dy dx = \int_{x=0}^{x=1} \left(\int_{y=0}^{y=1} \left[xz + yz + \frac{z^2}{2} \right]_0^1 dy \right) dx$ **/1 mark**

$$= \int_{x=0}^{x=1} \left(\int_{y=0}^{y=1} \left(x + y + \frac{1}{2} \right) dy \right) dx = \int_{x=0}^{x=1} \left[xy + \frac{y^2}{2} + \frac{1}{2}y \right]_0^1 dx$$

/1 mark

$$= \int_{x=0}^{x=1} x + 1 dx = \left[\frac{x^2}{2} + x \right]_0^1 = \frac{3}{2}$$

/1 mark

g) $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left(xy + \frac{x}{y} \right) = x - \frac{x}{y^2}$ **/0.5 marks**

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(x - \frac{x}{y^2} \right) = \frac{2x}{y^3} \Rightarrow y \frac{\partial^2 u}{\partial y^2} = \frac{2x}{y^2} \quad /1 \text{ mark}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(x - \frac{x}{y^2} \right) = 1 - \frac{1}{y^2} \Rightarrow 2x \frac{\partial^2 u}{\partial x \partial y} = 2x \left(1 - \frac{1}{y^2} \right) = 2x - \frac{2x}{y^2} \quad /1 \text{ mark}$$

$$y \frac{\partial^2 u}{\partial y^2} + 2x \frac{\partial^2 u}{\partial x \partial y} = \frac{2x}{y^2} + 2x - \frac{2x}{y^2} = 2x \quad /0.5 \text{ marks}$$

$$\text{Hence, } y \frac{\partial^2 u}{\partial y^2} + 2x \frac{\partial^2 u}{\partial x \partial y} = 2x$$

SECTION B: ATTEMPT ONLY TWO (2) QUESTIONS /30 marks

Question two.

a) Evaluate $\int_C xy^4 ds$, C is the right half of the circle $x^2 + y^2 = 16$ rotated in the counter clockwise direction. **(8 marks)**

b) Use the change of variables for evaluating the following double integral: **(7 marks)**

$$\iint_D \ln(x^2 + y^2) dA, \text{ where } D = \{(x, y) / x^2 + y^2 = e \text{ and } x^2 + y^2 = 4\}$$

Answers:

a) Since the parametric equation of the circle is

$$\begin{cases} x = 4 \cos t \\ y = 4 \sin t \end{cases}, \quad /1 \text{ mark}$$

$$\Rightarrow \begin{cases} \frac{dx}{dt} = -4 \sin t \\ \frac{dy}{dt} = 4 \cos t \end{cases}, \quad -\frac{\pi}{2} < t < \frac{\pi}{2} \quad /2 \text{ marks}$$

$$\text{Therefore } ds = \sqrt{(-4 \sin t)^2 + (4 \cos t)^2} dt = 4 dt. \quad /1 \text{ mark}$$

The line integral is then

$$\int_C xy^4 ds = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4(4\cos t)(4\sin t)^4(4) dt \quad /1 \text{ mark}$$

$$= 4096 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos t \sin^4 t dt \quad /1 \text{ mark}$$

$$= \frac{4096}{5} \left[\sin^5 \frac{\pi}{2} - \sin^5 \left(-\frac{\pi}{2} \right) \right] = \frac{4096}{5} \times 2 = \frac{8192}{5}. \quad /2 \text{ marks}$$

b) In polar coordinate we have:

$$\begin{cases} x = r \cos \alpha \\ y = r \sin \alpha \end{cases} \Rightarrow x^2 + y^2 = r^2 \text{ and } \sqrt{e} \leq r \leq 2 \text{ and } 0 \leq \alpha \leq 2\pi \quad /2 \text{ marks}$$

$$\iint_D \ln(x^2 + y^2) dA \text{ where } D = \{(x, y) / x^2 + y^2 = e \text{ and } x^2 + y^2 = 4\}$$

In polar coordinate this integral becomes:

$$\int_{\sqrt{e}}^2 \left(\int_0^{2\pi} r \ln r d\alpha \right) dr = \int_{\sqrt{e}}^2 (r \ln r [\alpha]_0^{2\pi}) dr = 2\pi \int_{\sqrt{e}}^2 (r \ln r) dr \quad /2 \text{ marks}$$

Using integration by part, we get:

$$= \pi \left\{ [r^2 \ln r]_{\sqrt{e}}^2 - \int_{\sqrt{e}}^2 r dr \right\} \quad /2 \text{ marks}$$

$$= \pi \left\{ \ln 16 - e \ln \sqrt{e} - 2 + \frac{e}{2} \right\} \quad /1 \text{ mark}$$

Question three.

a) Find the general solution of the following homogeneous linear system, using

$$\text{the vector- matrix methods, } \begin{cases} x'_1 = x_1 + x_2 - x_3 \\ x'_2 = 2x_1 + 3x_2 - 4x_3. \\ x'_3 = 4x_1 + x_2 - 4x_3 \end{cases} \quad (9 \text{ marks})$$

b) Evaluate $\iint_S u(x, y, z) ds$ where S is the surface of the paraboloid

$$z = 2 - (x^2 + y^2) \text{ above the } xy\text{-plane and } u(x, y, z) \text{ equal to } 1 \quad (6 \text{ marks})$$

Answers:

a) Assuming a solution of this is of the form: $x = \beta e^{\lambda t}$ that is:

$$x_1 = \beta_1 e^{\lambda_1 t}, x_2 = \beta_2 e^{\lambda_2 t}, x_3 = \beta_3 e^{\lambda_3 t}$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & -4 \\ 4 & 1 & -4 \end{pmatrix} \rightarrow |A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & -1 \\ 2 & 3-\lambda & -4 \\ 4 & 1 & -4-\lambda \end{vmatrix} = 0 \quad \textbf{/0.5 marks}$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = -3 \quad \textbf{/1.5 marks}$$

$$A\beta = \lambda\beta$$

Let consider $\lambda = \lambda_1 = 1$, then we get:

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & -4 \\ 4 & 1 & -4 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{1} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \Rightarrow \beta_1 = 1 = \beta_2 = \beta_3 \quad \textbf{/ 0.5 marks}$$

$$\beta^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \textbf{/ 0.5 marks}$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t \Leftrightarrow x = \begin{pmatrix} e^t \\ e^t \\ e^t \end{pmatrix} \quad \textbf{/ 0.5 marks}$$

Let consider $\lambda_2 = 2$, then we get:

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & -4 \\ 4 & 1 & -4 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{2} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \Rightarrow \beta_1 = \beta_3 = 1, \beta_2 = 2 \quad \textbf{/ 0.5 marks}$$

$$\text{and } \beta^{(2)} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad \textbf{/ 0.5 marks}$$

$$x = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} e^{2t} \Leftrightarrow x = \begin{pmatrix} e^{2t} \\ 2e^{2t} \\ e^{2t} \end{pmatrix} \quad \textbf{/ 0.5 marks}$$

Let consider $\lambda_3 = -3$, then we get:

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & -4 \\ 4 & 1 & -4 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{-3} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \Rightarrow \beta_1 = 1 \Rightarrow \beta_3 = 11, \beta_2 = 7 \quad \textbf{/ 0.5 marks}$$

$$\text{and } \beta^{(3)} = \begin{pmatrix} 1 \\ 7 \\ 11 \end{pmatrix} \quad \textbf{/ 0.5 marks}$$

$$x = \begin{pmatrix} 1 \\ 7 \\ 11 \end{pmatrix} e^{-3t} \Leftrightarrow x = \begin{pmatrix} e^{-3t} \\ 7e^{-3t} \\ 11e^{-3t} \end{pmatrix} \quad / \text{ 0.5 marks}$$

If those three solution are linearly independent, then we have:

$$x = c_1 \begin{pmatrix} 3e^t \\ 2e^t \\ 2e^t \end{pmatrix} + c_2 \begin{pmatrix} e^{2t} \\ 2e^{2t} \\ e^{2t} \end{pmatrix} + c_3 \begin{pmatrix} e^{-3t} \\ 7e^{-3t} \\ 11e^{-3t} \end{pmatrix} \quad / \text{ 1 mark}$$

Where c_1, c_2, c_3 are arbitrary constants.

$$\text{The general solution is: } \begin{cases} x_1 = 3c_1 e^t + c_2 e^{2t} + c_3 e^{-3t} \\ x_2 = 2c_1 e^{2t} + 2c_2 e^{2t} + 7c_3 e^{-3t} \\ x_3 = 2c_1 e^{2t} + c_2 e^{2t} + 11c_3 e^{-3t} \end{cases} \quad / \text{ 1.5 marks}$$

Where c_1, c_2, c_3 are arbitrary constants.

$$\text{b) If then } u(x, y, z) = 1, \text{ then } \iint_S u(x, y, z) ds = \iint_D \sqrt{1+4x^2+4y^2} dx dy \quad / \text{ 1 mark}$$

Passing to polar coordinates:

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \text{ where } 0 \leq r \leq \sqrt{2} \text{ and } 0 \leq \theta \leq 2\pi \quad / \text{ 1 mark}$$

$$\iint_D \sqrt{1+4x^2+4y^2} dx dy = \int_0^{2\pi} \left(\int_0^{\sqrt{2}} r \sqrt{1+4r^2} dr \right) d\theta \quad / \text{ 1 mark}$$

Calculating $\int_0^{\sqrt{2}} r \sqrt{1+4r^2} dr$ by change of variable:

$$u = 1+4r^2 \Rightarrow du = 8r dr \Rightarrow r dr = \frac{u}{8} du \quad / \text{ 0.5 marks}$$

$$\text{If } r=0 \Rightarrow u=1, r=\sqrt{2} \Rightarrow u=9$$

$$\int_0^{\sqrt{2}} r \sqrt{1+4r^2} dr = \frac{1}{8} \int_1^9 \sqrt{u} du = \frac{1}{12} \left[u^{3/2} \right]_1^9 = \frac{1}{12} \left[9^{3/2} - 1^{3/2} \right] = \frac{13}{6} \quad / \text{ 1 mark}$$

$$\int_0^{2\pi} \left(\int_0^{\sqrt{2}} r \sqrt{1+4r^2} dr \right) d\theta = \int_0^{2\pi} \left(\frac{13}{6} \right) d\theta = \frac{13}{6} [\theta]_0^{2\pi} \quad / \text{ 0.5 marks}$$

$$= \frac{13 \times 2\pi}{6} = \frac{13\pi}{3}$$

/ 1 mark

Question four.a) Evaluate $\int_{(1,2)}^{(3,4)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy$

(7 marks)

b)

i. Evaluate $\oint_C xydx + x^2dy$ around the sides of the square with vertices $D(0,0), E(1,0), F(1,1)$ and $G(0,1)$.

(5 marks)

ii. Convert the line integral to a double integral and verify Green's theorem.

(3 marks)

Answer:

a) $\frac{\partial L}{\partial y} = 12xy - 3y^2 = \frac{\partial M}{\partial x}$, so there exists a function $f(x, y)$ such that $Ldx + Mdy = df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$.

/ 1 mark

$$\frac{\partial f}{\partial x} = 6xy^2 - y^3, \frac{\partial f}{\partial y} = 6x^2y - 3xy^2$$

/ 1 mark

$$f(x, y) = 3x^2y^2 - xy^3 + g(y)$$

/ 1 mark

$$\begin{cases} \frac{\partial f}{\partial y} = 6x^2y - 3xy^2 + g'(y) \\ \frac{\partial f}{\partial y} = 6x^2y - 3xy^2 \end{cases} \rightarrow g'(y) = 0 \rightarrow g(y) = c$$

/ 1 mark

$$f(x, y) = 3x^2y^2 - 3xy^2 + c$$

/ 1 mark

$$\int_{(1,2)}^{(3,4)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy = f(x, y)|_{(1,2)}^{(3,4)} = f(3,4) - f(1,2) \text{ where } c=0$$

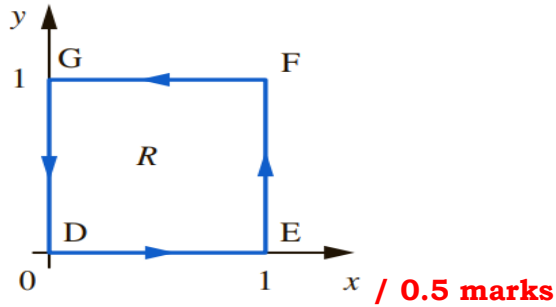
/ 1 mark

$$= (3.9.16 - 4^3.3) - (3.1.2^2 - 2^3.1) = 236$$

/ 1 mark

b)

- i. The path of integration is shown in Figure below. To apply Green's theorem the path of integration should be followed in such a way that the region of integration is always to its left. We must therefore travel anticlockwise around C.



/ 0.5 marks

On DE, $y = 0 \Rightarrow dy = 0$ and $0 \leq x \leq 1$. / 0.5 marks

On EF, $x = 1 \Rightarrow dx = 0$ and $0 \leq y \leq 1$. / 0.5 marks

On FG, $y = 1 \Rightarrow dy = 0$ and x decreases from 1 to 0. / 0.5 marks

On GD, $x = 0 \Rightarrow dx = 0$ and y decreases from 1 to 0. / 0.5 marks

The integral around the curve C can then be written as

$$\oint_C = \int_D^E + \int_E^F + \int_F^G + \int_G^D \quad / 0.5 \text{ marks}$$

Therefore

$$\oint_C xydx + x^2dy = 0 + \int_0^1 1dy + \int_1^0 xdx + 0 \quad / 1 \text{ mark}$$

$$= [y]_0^1 + \left[\frac{x^2}{2}\right]_1^0 \quad / 0.5 \text{ marks}$$

$$= 1 - \frac{1}{2} = \frac{1}{2} \quad / 0.5 \text{ marks}$$

$$\begin{aligned} \text{ii.} \quad \oint_C xydx + x^2dy &= \iint_R (2x - x) dx dy \quad / 1 \text{ mark} \\ &= \int_0^1 \int_0^1 (2x - x) dx dy \quad / 1 \text{ mark} \\ &= \int_0^1 \left[\frac{x^2}{2}\right]_1^0 dy = \int_0^1 \frac{1}{2} dy = \frac{1}{2} \quad / 1 \text{ mark} \end{aligned}$$